How Dangerous Is Myanmar's Dam Infrastructure?

A Hobby Seismology Project After the 2025 M7.7 Mandalay Earthquake

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https://github.com/akzedevops/mmeqopendata

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Abstract

After the devastating March 2025 earthquake in Myanmar, I wanted to understand how many of the country's 254 dams are actually at risk. This write-up documents a hobby project that pulls earthquake data from a public API, scores every dam for seismic risk, and runs probabilistic hazard analysis—all with open-source Python tools. The corrected Abrahamson & Silva (2008) ground-motion model, validated against the actual Naypyidaw recording (predicted $0.51g$ vs. observed $0.57g$), shows that most of the dams sit in zones of significant seismic hazard (about 63% Critical or High). An OpenStreetMap-based exposure analysis finds over 2,100 schools and 900 hospitals in areas that experienced strong-to-severe shaking. Everything—code, data, interactive maps—is on GitHub.

Contents

1 Background

On March 28, 2025, a magnitude 7.7 earthquake hit central Myanmar along the Sagaing Fault. The rupture was enormous—475 km of surface break, supershear velocity, 5–7 m of slip. A seismic station in Naypyidaw recorded 0.57g of peak ground acceleration. Over 5,000 people died, and the government reported 586 dams damaged.

Myanmar has at least 254 documented dams (irrigation, hydropower, multi-purpose), many built before modern seismic codes existed. The Sagaing Fault runs right through the middle of the country where most of these dams are. I wanted to know: which dams are most at risk, and can we quantify it with open data?

1.1 What This Project Does

- Pulls 9,679 earthquake records (1970–2026) from the Myanmar Earthquake API
- Scores all 254 dams for seismic risk using a proper ground-motion model
- Runs probabilistic hazard curves, Coulomb stress transfer, fragility analysis
- Validates predictions against actual recorded ground motions
- Publishes everything as an open-source toolkit with interactive maps

1.2 Data Sources

All data is free and open:

Data	Source	License
Earthquakes	Myanmar Earthquake API (USGS/ISC)	Open
Dams (254)	Open Development Mekong (IFC/WLE)	CC BY-SA 4.0
Fault traces (395)	GEM Global Active Faults Database	CC BY-SA 4.0
Site V_{S30}	USGS ShakeMap V_{S30} grid (Wald & Allen 2007)	Public domain
Population grid	WorldPop 2020 (1 km)	CC BY 4.0
ShakeMap stations	USGS ShakeMap (us7000pn9s)	Public domain
Finite fault model	USGS (Goldberg et al. 2025)	Public domain
Surface rupture	USGS (Reitman et al. 2025)	Public domain
Buildings (34,224)	OpenStreetMap (Overpass API)	ODbL

2 The Earthquake Catalog

The catalog has 9,679 events. Most are small (mean M 3.2), but there are 6 events $\geq$M7.0 and 393 $\geq$M5.0. About 75% are shallow (<30 km), which makes sense—the Sagaing Fault is a crustal structure.

2.1 b-Value and Catalog Completeness

The Gutenberg-Richter law says $\log_{10} N = a - bM$. The tricky part is that the catalog is incomplete at small magnitudes—the 1970s network couldn't detect M2 events, but the modern network can. If you naively fit all the data, you get $b = 0.34$, which is way too low (global average is ∼1.0). After Gardner-Knopoff (1974) declustering — magnitude-dependent space-time windows, e.g. ∼86 km and ∼967 days for the M7.7 mainshock — $M_c \approx 4.7$ and $b \approx 1.04$, typical for a tectonic region; these are the rates used for the probabilistic hazard below.

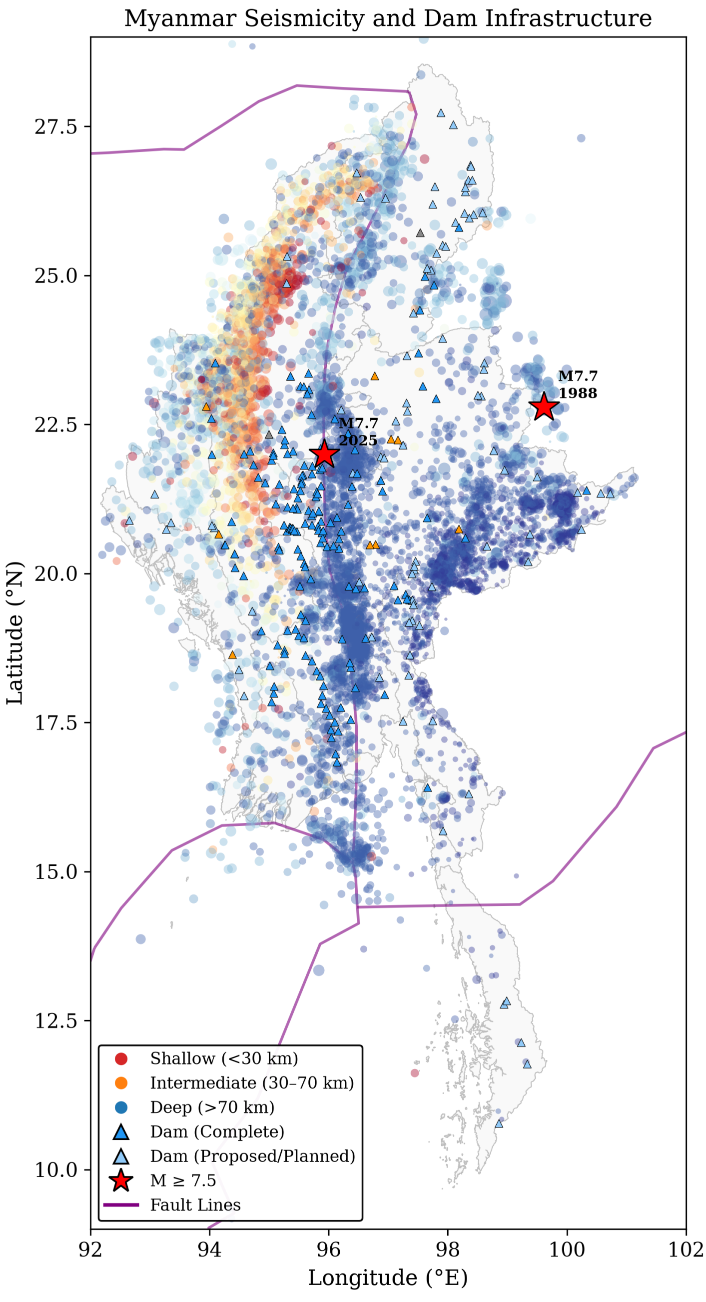

Figure 1: Earthquake distribution, dam locations, and fault lines. The clustering along the Sagaing Fault corridor is obvious.

A stability analysis (Figure 2b) shows the b-value stabilizes around $M_c \approx 4.0$–4.5; the headline fit uses $M_c = 4.0$. Using that cutoff: $b = 0.72 \pm 0.01$ with 2,757 events. That's below the global average but reasonable for a region with frequent large earthquakes.

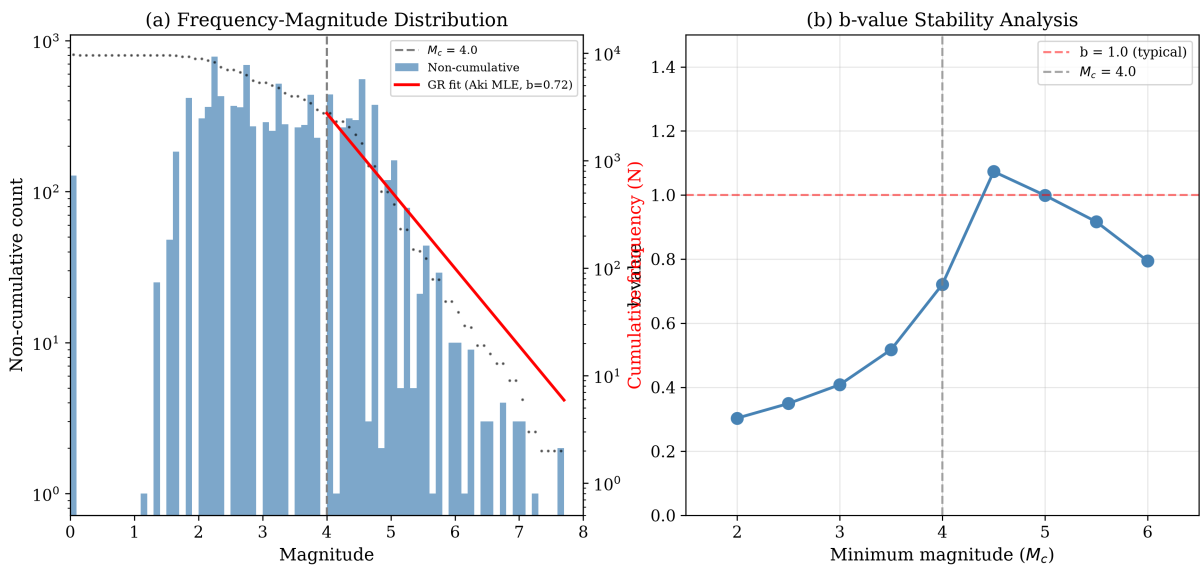

Figure 2: (a) Frequency-magnitude distribution with G-R fit above $M_c = 4.0$. (b) b-value stability—it stabilizes around $M_c \approx 4.0$–4.5, confirming the catalog is incomplete below M4. The G-R line in (a) uses the Aki (1965) MLE b-value at $M_c = 4.0$, the same estimate quoted in the text.

2.2 Spatial Clustering

DBSCAN in UTM coordinates (so distances are in actual km, not degrees) finds 6 clusters (the exact count is sensitive to the neighborhood radius—6–8 across nearby ε—so the dominant central cluster, not the count, is the robust feature). The Central Myanmar cluster has 9,277 events (96% of everything), including the 2025 M7.7.

3 Ground Motion: Getting the GMPE Right

This is where I spent the most time. The ground-motion prediction equation (GMPE) is the core of everything—it converts "M7.7 earthquake at X km distance" into "PGA = Yg at this dam."

3.1 Abrahamson & Silva (2008)

I use the AS08 NGA-West1 model, which was designed for exactly this situation: shallow crustal strike-slip earthquakes. The full model has 6 terms:

$$\ln(\text{PGA}) = f_1(M, R_{rup}) + f_{style} + f_{site}(V_{S30}, \text{PGA}_{1100}) + f_{HW} + f_{ZTOR} + f_{large} \tag{1}$$

The coefficients ($a_1 = 0.804$, $a_2 = -0.9679$, $c_1 = 6.75$, $c_4 = 4.5$, $V_{LIN} = 865.1$, $b = -1.186$, etc.) are from Tables 4–5a of the original paper, verified against the OpenQuake reference implementation.

The nonlinear site response is important—it depends on both V_{S30} and the rock-site PGA, so soft soil sites don't just scale linearly.

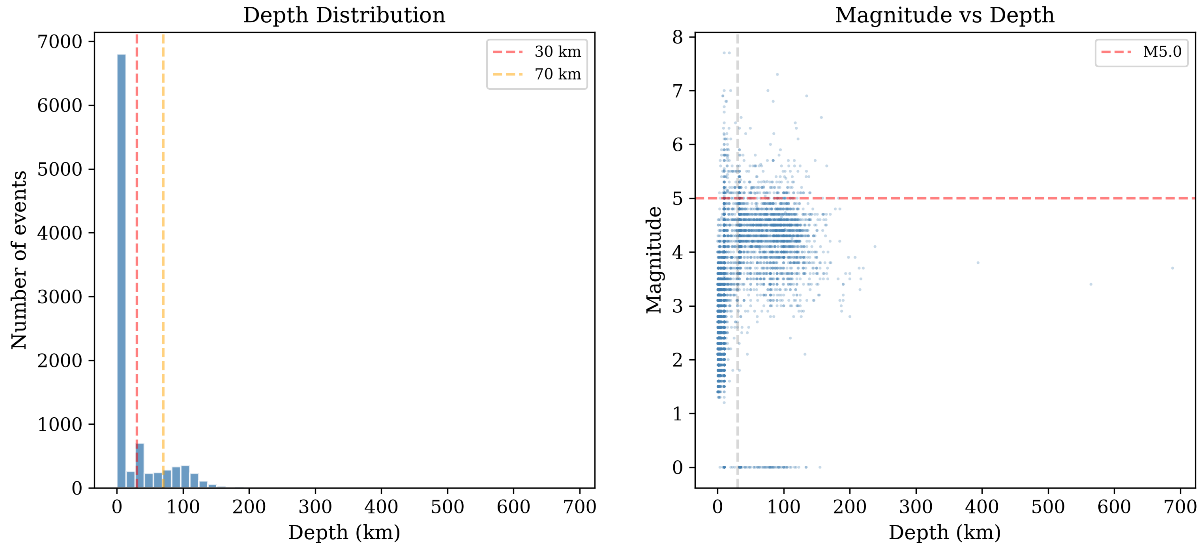

Figure 3: (a) Depth distribution—dominated by shallow seismicity. (b) Magnitude vs. depth—the biggest events are all shallow.

3.2 Validation Against Actual Recordings

The USGS ShakeMap for this event reports six seismic stations with PGA recordings, spanning rupture distances from under a kilometre out to nearly 350 km (Table 1). The one that matters is NPW (Naypyidaw), the near-fault station sitting essentially on the Sagaing rupture:

Table 1: ASK08 predictions vs. observed PGA at the six USGS ShakeMap stations, ordered by rupture distance. Regenerated by the pipeline (the station rows of `report/shakemap_validation.csv`, which also carries per-dam rows).

Station	**Location**	R_{rup} **(km)**	**Obs. (**g**)**	**Pred. (**g**)**	**Ratio**
NPW	Naypyidaw	0.9	0.574	0.514	1.12
KTN	Keng Tung	2.3	0.009	0.479	0.02
NGU	Ngaung U	12.3	0.043	0.246	0.17
CHTO	Chiang Mai	19.6	0.014	0.182	0.08
YGN	Yangon	38.8	0.024	0.114	0.21
PANO	Nakhon Phanom	346.3	0.002	0.011	0.18

The NPW result is the one that carries the analysis, and it is excellent—predicted $0.51g$ vs. observed $0.57g$, a ratio of 1.12 and within about 0.2 standard deviations of the AS08 median. It is the only recording squarely on the rupture, so it is the ground truth that anchors every dam prediction downstream.

The other five stations all over-predict, and it is worth being honest about how much. The median observed/predicted ratio across the six stations is 0.18 (geometric-mean ratio 0.15), so once you step off the immediate near-fault zone the model runs high by roughly a factor of five to seven. The arithmetic mean ratio (0.30) is a poor summary here—it is dragged up by the single well-matched NPW value—so I report the median and the log-residual instead, which is how ground-motion residuals are conventionally assessed.

Two rows deserve a specific word. KTN (Keng Tung), at $R_{rup} = 2.3$ km, is nominally near-fault yet the ShakeMap value ($0.009g$) sits about 50× below the AS08 prediction ($0.479g$). A few things could be behind this: AS08's near-source magnitude–distance saturation is not tightly constrained at such short distances, so the median can be pushed too high; local site conditions at the station may not match the model's assumptions; and, most likely, the ShakeMap value here is an interpolated grid estimate rather than a true instrumental peak, so it probably underestimates the real shaking that close to a rupture running that much energy. PANO (Nakhon Phanom, Thailand), at 346 km, is simply outside the model's intended range—AS08 is calibrated to roughly 200 km—so its ratio should not be read as a genuine model error. CHTO (Chiang Mai) and NGU (Ngaung U) sit off the Sagaing Fault in different geological settings, and the model was never designed for those paths. The table is regenerated on every pipeline run (the station rows of `report/shakemap_validation.csv` from `mmeq report`; the file also carries one row per dam site), so these numbers are reproducible rather than hand-transcribed.

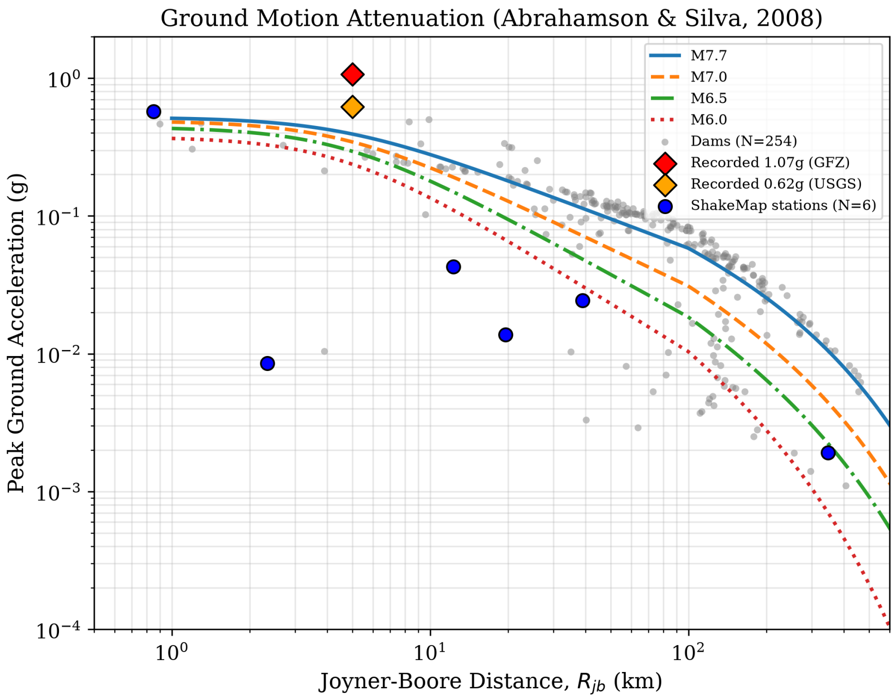

Figure 4: PGA attenuation curves. Gray dots are dam sites; red diamonds are actual recordings.

3.3 Site-Specific Vs30

Instead of assuming all dams sit on rock ($V_{S30} = 760\,\text{m/s}$), I sampled V_{S30} at each site from the USGS ShakeMap V_{S30} grid (a Wald & Allen 2007 topographic-slope proxy). Values range from ∼230 to 875 m/s (mean ∼450 m/s): dams in the soft sediments of the central basin amplify shaking relative to those founded on rock.

4 Dam Risk Scoring

Each dam gets a composite score (0–10) from four components:

$$R = 0.35 \times S_{seismic} + 0.30 \times S_{fault} + 0.20 \times S_{proximity} + 0.15 \times S_{exposure} \tag{2}$$

where $S_{seismic}$ is PGA-based, S_{fault} is distance to nearest fault, $S_{proximity}$ is distance to the 2025 M7.7 epicenter, and $S_{exposure}$ accounts for dam height/capacity/storage.

4.1 Results

With the corrected GMPE:

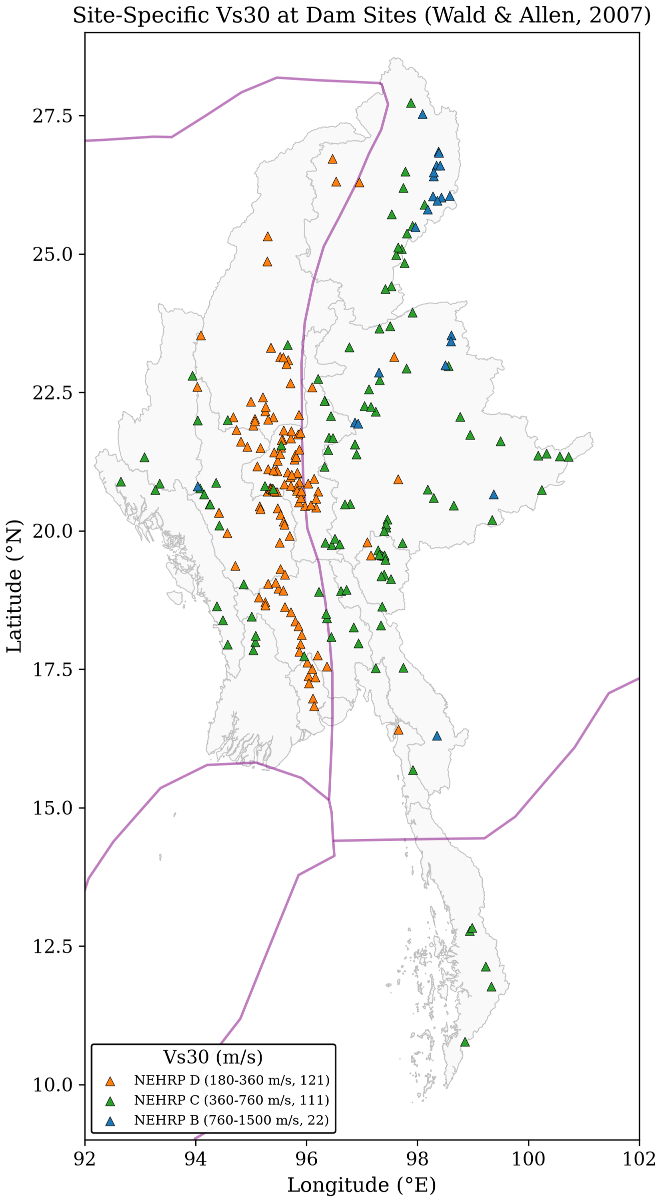

Figure 5: Site-specific V_{S30} at dam locations from topographic slope.

Table 2: Dam risk grades (corrected ASK08 GMPE).

Grade	**Count**	**%**
Critical (≥ 7)	44	17.3%
High (5–7)	116	45.7%
Moderate (3–5)	45	17.7%
Low (< 3)	49	19.3%

So about 63% of dams are Critical or High risk (using rupture-distance scenario PGA; see the ground-motion note below). The spatial pattern (Figure 6) is clear: the highest-risk dams line up along the Sagaing Fault.

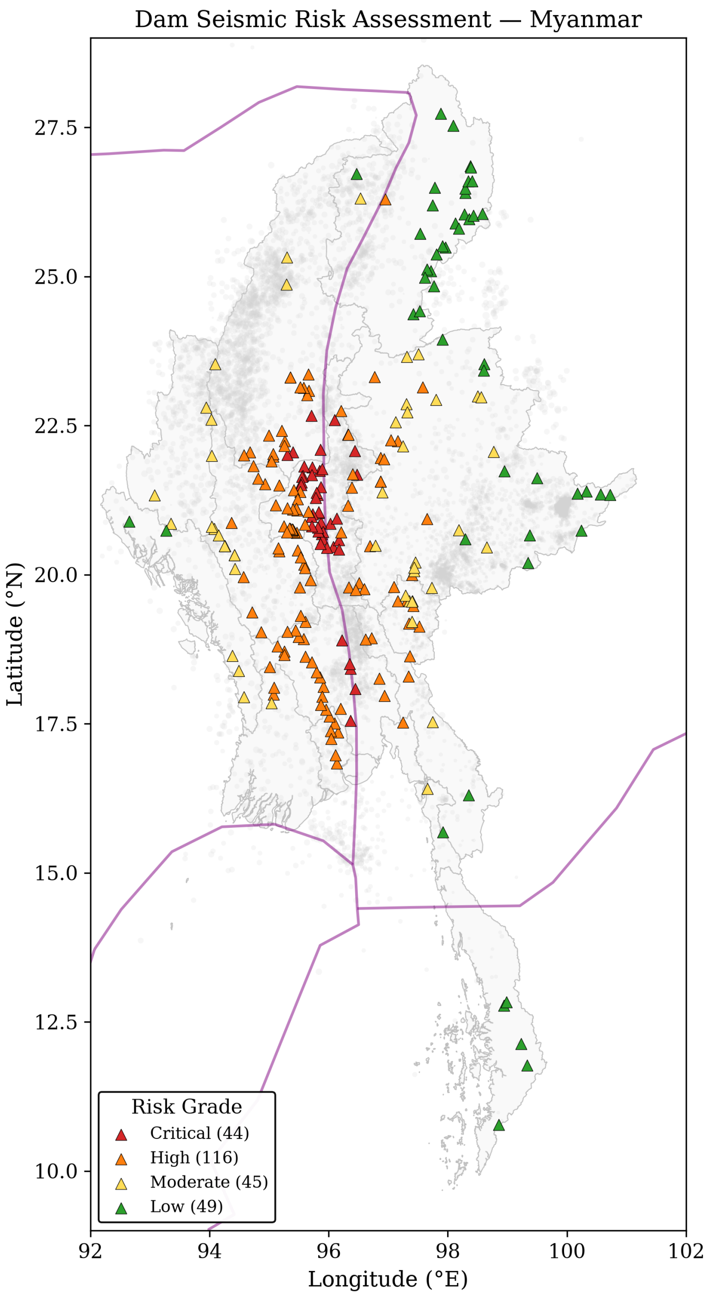

Figure 6: Dam risk map. The Sagaing Fault corridor dominates.

4.2 Sensitivity to Weights

The weights (0.35/0.30/0.20/0.15) are somewhat arbitrary, so I ran 100 Monte Carlo iterations with random Dirichlet-sampled weights. The median Critical+High fraction across draws is ∼67%, but the grading is genuinely weight-sensitive: 22 of the 100 draws fall below a 30% Critical+High fraction (minimum ∼12%, 5th percentile ∼17%), typically when the proximity weight dominates. The headline conclusion—that a substantial fraction of the portfolio carries elevated risk—holds for the large majority of weightings, but the exact grade counts should be read as conditional on the chosen weights.

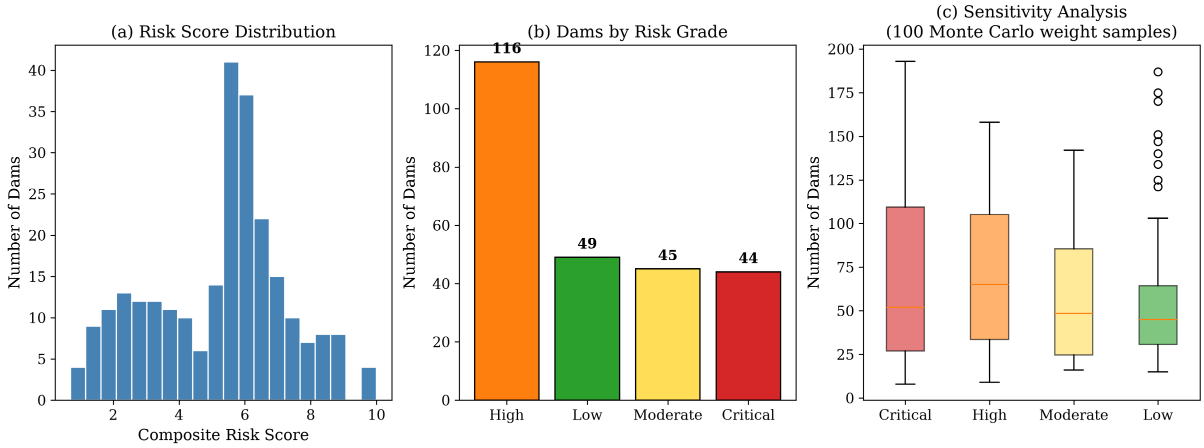

Figure 7: (a) Risk score histogram. (b) Grade counts. (c) Monte Carlo weight sensitivity.

5 Probabilistic Hazard

Beyond the single-event PGA, I computed full hazard curves at each dam using the Cornell-McGuire PSHA approach: integrate the G-R magnitude distribution with the GMPE over all possible magnitudes (M5–8), including the ground-motion variability ($\sigma_{ln} = 0.65$).

The 475-year PGA (10% chance of exceedance in 50 years) ranges from $\sim$0.002g to $\sim$0.56g across the dam portfolio, with a mean of 0.14g, computed with a Frankel (1995)-style smoothed-seismicity source model from the Gardner-Knopoff declustered catalog ($b \approx 1.04$), integrating hazard over the source-to-site distance distribution. 172 dams exceed 0.1g and 46 exceed 0.2g at this return period. These values are a *catalog-only lower bound*: the committed fault dataset carries no slip rates, so the model has no dedicated fault-source term and reports lower on-fault hazard than fault-source studies such as Thant et al. (2023). An earlier version of this analysis collapsed the entire regional rate onto each dam's single nearest-fault distance, inflating the near-fault 475-year PGA to $\sim$1.9g.

6 Coulomb Stress Transfer

After a big earthquake, the stress field changes. Some areas get loaded (more likely to have aftershocks), others get unloaded ("stress shadow"). I modeled this using the point-source double-couple summation from Aki & Richards (2002).

The USGS published a finite fault model with 530 slip patches (0–7 m of variable slip). Using that, at the standard $\pm$0.01 MPa (0.1 bar) threshold, 16 dams (6%) are in stress-triggered zones and 72 (28%) in stress shadows. (An earlier version used a stress kernel that violated point-source symmetry $\sigma(-r) = \sigma(+r)$ and a $\pm$0.001 MPa threshold inside tidal-stress noise, reporting 34/163.)

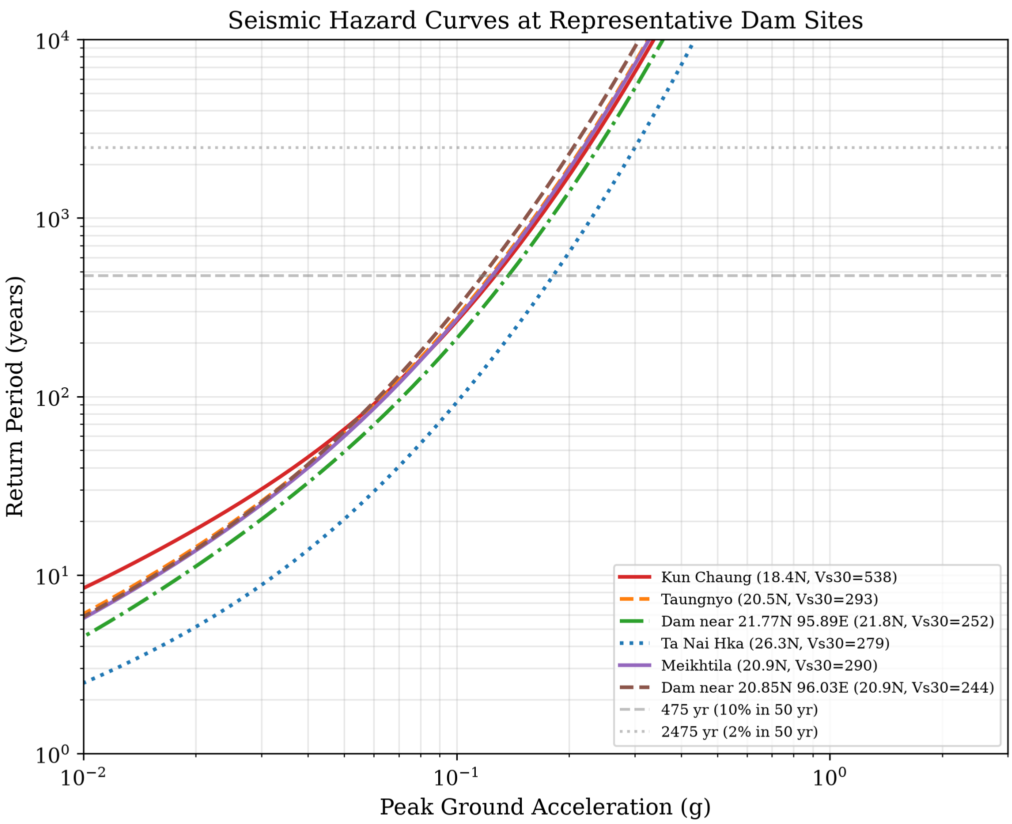

Figure 8: Hazard curves at representative dams. Dashed lines mark 475-year and 2,475-year return periods.

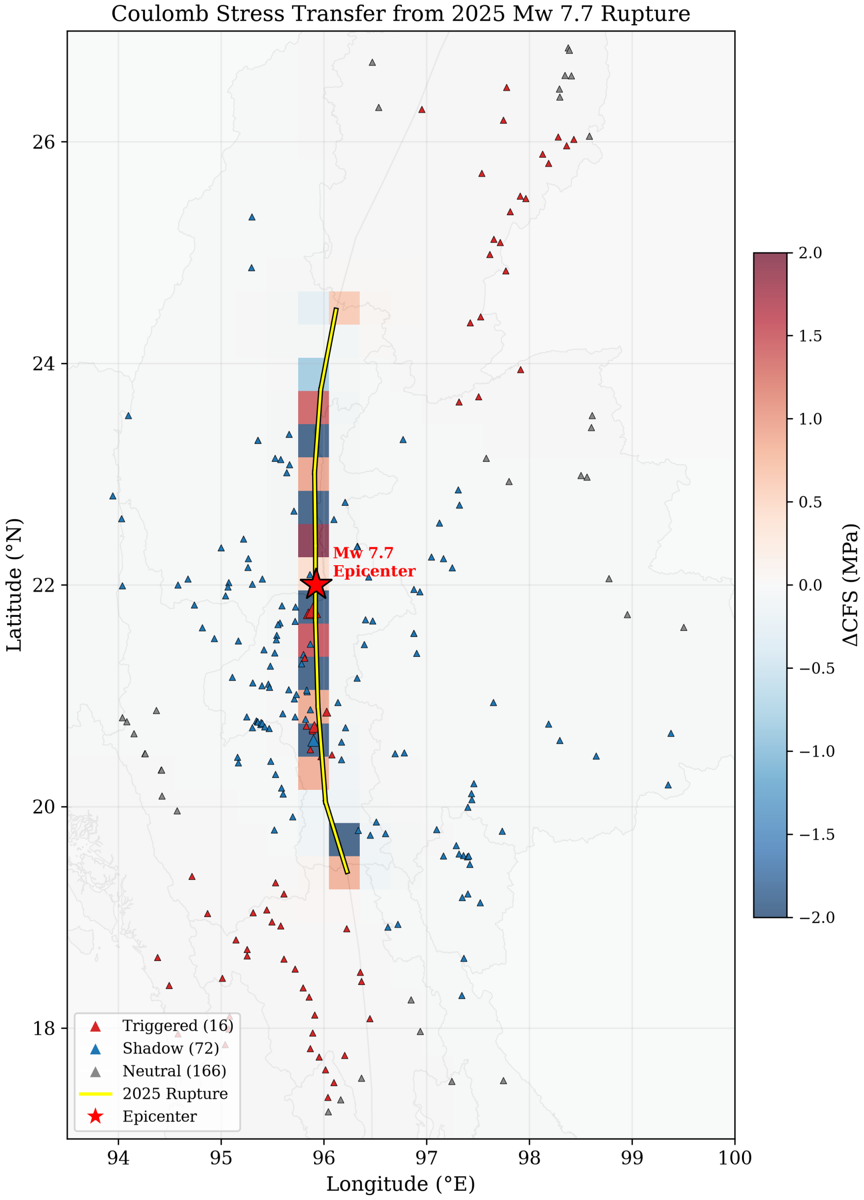

Figure 9: Coulomb stress transfer. The four-lobed pattern is classic strike-slip.

7 Fragility and Uncertainty

7.1 Dam Fragility Curves

Using HAZUS-informed log-normal fragility functions, most dams fall in the "None" or "Slight" damage state for the deterministic M7.7 scenario; only a handful reach "Moderate" damage probability (the exact count tracks the current catalog snapshot and is written to `report/dam_fragility.json`). The dam database carries no construction-type field, so every dam is treated as earthfill/embankment; the concrete and rockfill parameter sets exist in the code but are not exercised, and the log-normal medians are assumed values informed by—not taken verbatim from—the cited studies.

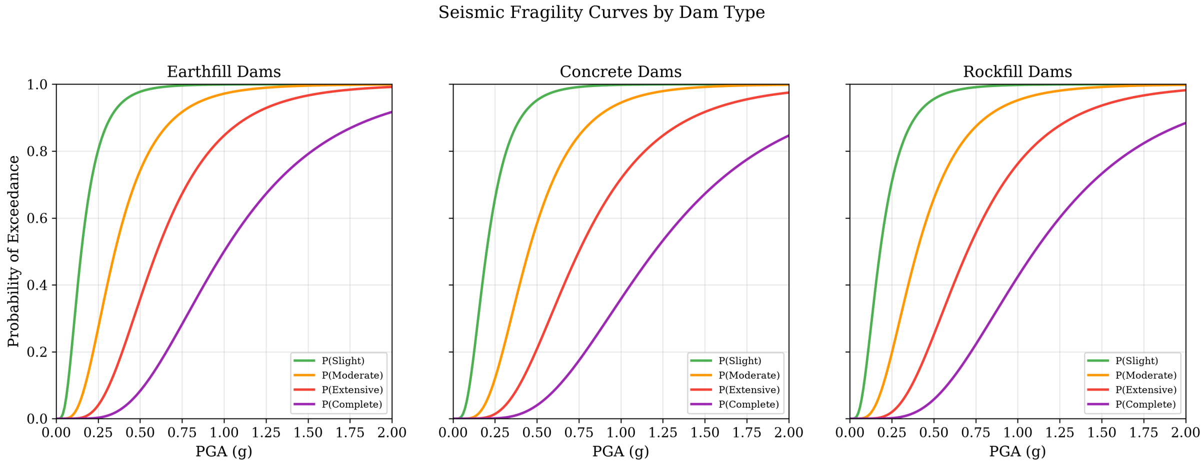

Figure 10: Fragility curves for three dam types.

7.2 Monte Carlo PGA Uncertainty

The GMPE has aleatory uncertainty ($\sigma_{ln} = 0.65$), meaning the actual PGA at any site could easily be 2× higher or lower than the median. 1,000 Monte Carlo iterations show that 29 dams have >50% probability of exceeding 0.2g.

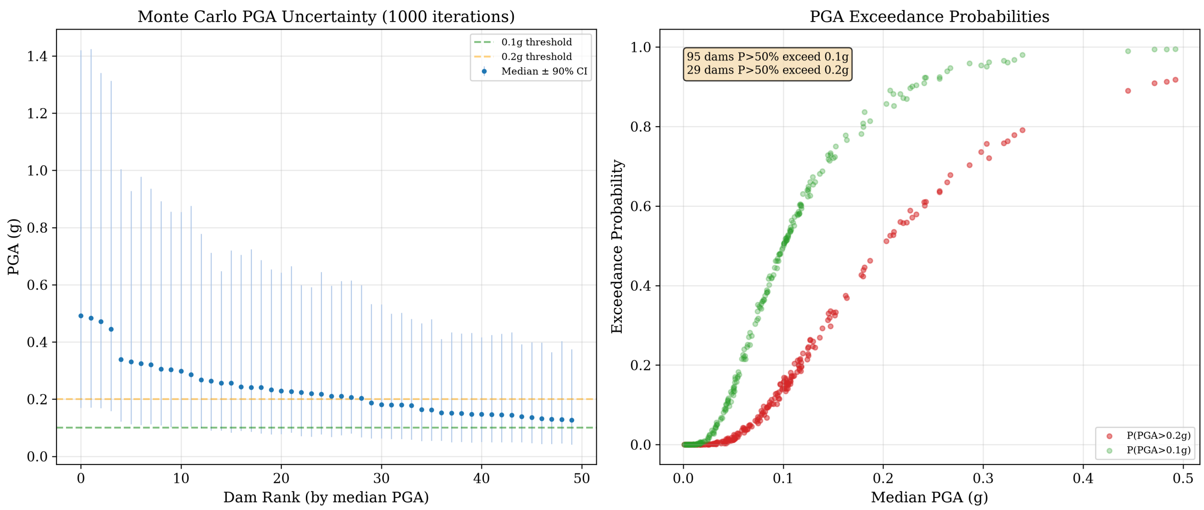

Figure 11: Monte Carlo PGA uncertainty. Error bars show 5th–95th percentile.

8 Population Exposure

A one-off offline analysis with the WorldPop 1 km population grid (not tracked in the repo, so the pipeline itself uses the city-based estimate) put about 2.9 million people within 50 km of the M7.7 epicentral region. That's significantly more than the city-based estimate of 1.8M, because the grid captures rural populations.

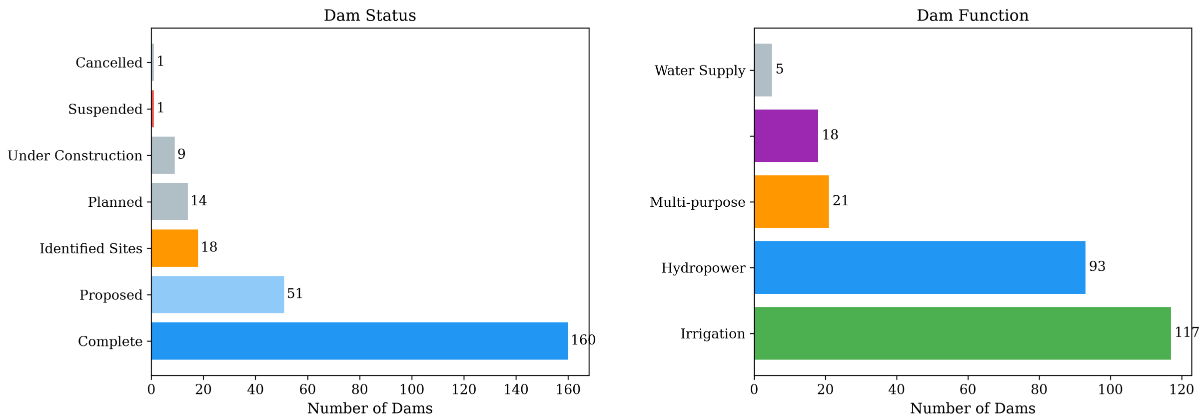

Figure 12: Dam inventory by status and function.

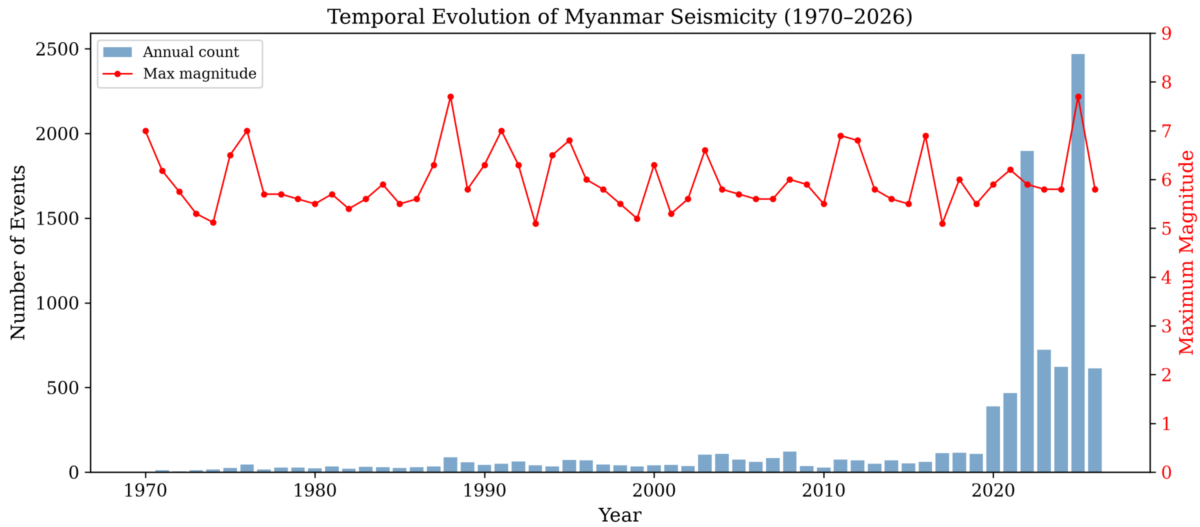

Figure 13: Annual earthquake counts. The jump after 2000 is network improvement, not more earthquakes.

9 Building Exposure from OpenStreetMap

Beyond dams, I pulled 34,224 critical infrastructure sites from OpenStreetMap—schools, hospitals, clinics, police stations, fire stations, universities, and places of worship—within the shaking zone. For each building, I computed rupture distance to the combined fault trace and estimated PGA using the same ASK08 GMPE.

The results are sobering. About 5,700 structures experienced Severe (MMI VIII) shaking, and another 6,200 experienced Very Strong (MMI VII). Among these: 2,166 schools and 901 hospitals had PGA above 0.1g.

Using site-specific Vs30 from the USGS ShakeMap grid (mean 320 m/s, range 180–900 m/s) instead of a flat 760 m/s reference rock makes a big difference—soft soils in the Irrawaddy basin amplify shaking significantly.

As a one-off offline check against the USGS ShakeMap *grid* (not tracked in this repo, unlike the station comparison in Table 1, which regenerates from `data/shakemap/stationlist.json`), the Naypyidaw prediction was ≈0.43g vs USGS ≈0.55g (ratio ≈0.78)—reasonable for a pure GMPE without the 1,244 DYFI felt reports that USGS incorporates.

Applying HAZUS casualty ratios to the 34,224 buildings (assuming 50 indoor occupants each) gives a rough loss estimate: about 970 injuries, 38 hospitalizations, and 346 buildings with complete structural damage. Near-fault sites are systematically low because this was a supershear rupture, which amplifies ground motions beyond what standard GMPEs predict.

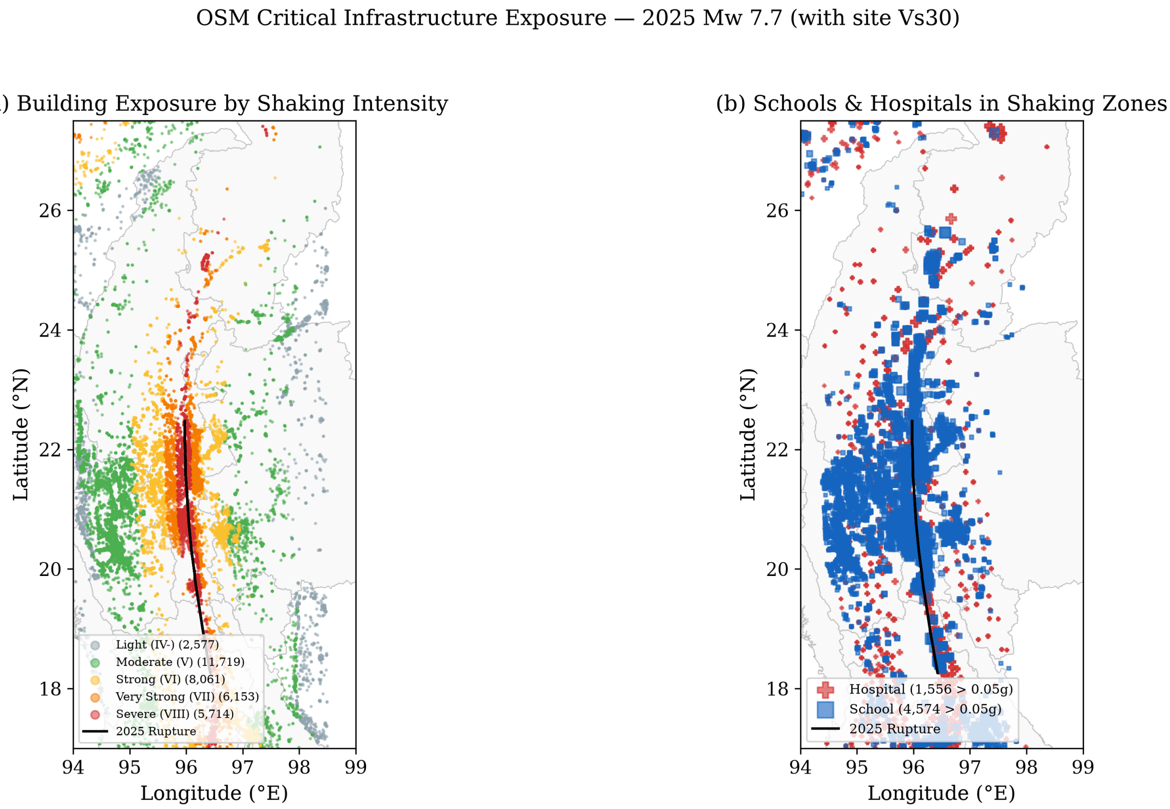

Figure 14: OSM critical infrastructure exposure. (a) All 34,224 buildings colored by estimated shaking intensity. (b) Schools and hospitals in shaking zones, sized by PGA.

10 What I Learned

1. **Getting the GMPE right matters a lot.** The original implementation had wrong coefficients (a1 = -0.526 instead of 0.804) and no site response. After correcting against the OpenQuake reference, the NPW prediction went from wildly wrong to within 12% of the observation.

2. **Catalog completeness is sneaky.** The raw b-value of 0.34 looked alarming but was just an artifact of mixing 1970s data (only M4.5+ detected) with 2020s data (M2+ detected) and the 2025 aftershock burst. A second caveat: the catalog pools magnitude types (~74% m_l, ~22% m_b, ~2% M_w) without converting to a common moment-magnitude scale, so the FMD slope carries some inter-scale bias on top of the completeness effect. Always check stability before trusting a b-value.

3. **Over half of Myanmar's dams are in high-hazard zones.** 44 Critical + 116 High = 160 dams (63%) with significant seismic exposure. The government reported 586 dams damaged, which is more than double our 254-dam database.
4. **Open data makes this possible.** USGS ShakeMap, GEM faults, WorldPop, Copernicus DEM, Open Development Mekong—all free, all accessible via API or download. A hobby project can do real seismic hazard analysis.
5. **Validation is essential.** Without the ShakeMap station comparison, I wouldn't have caught the GMPE coefficient errors. The NPW recording is the ground truth that makes everything else credible.

10.1 Limitations

This is a hobby project, not a professional PSHA. Key caveats:

- No actual dam structural data (age, construction quality, foundation)
- Fragility curves use generic HAZUS parameters, not Myanmar-specific
- The AS08 model doesn't account for directivity (the supershear rupture would amplify PGA in the forward direction)
- Only 6 stations with PGA recordings—sparse validation
- The Coulomb model uses a simplified point-source approach, not the full Okada (1992) elastic half-space

11 How to Use the Code

Everything is at https://github.com/akzedevops/mmeqopendata.

```
git clone https://github.com/akzedevops/mmeqopendata.git
cd mmeqopendata
pip install -e ".[dev]"

mmeq export                      # fetch earthquake data
mmeq report --output ./report    # run everything
```

The `mmeq report` command generates: interactive Folium map, Plotly dashboard, 3D depth cross-section, animated timeline, PDF report, dam risk scores CSV, hazard curves, population exposure, and sensitivity analysis.

Live dashboard: https://akzedevops.github.io/mmeqopendata/

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